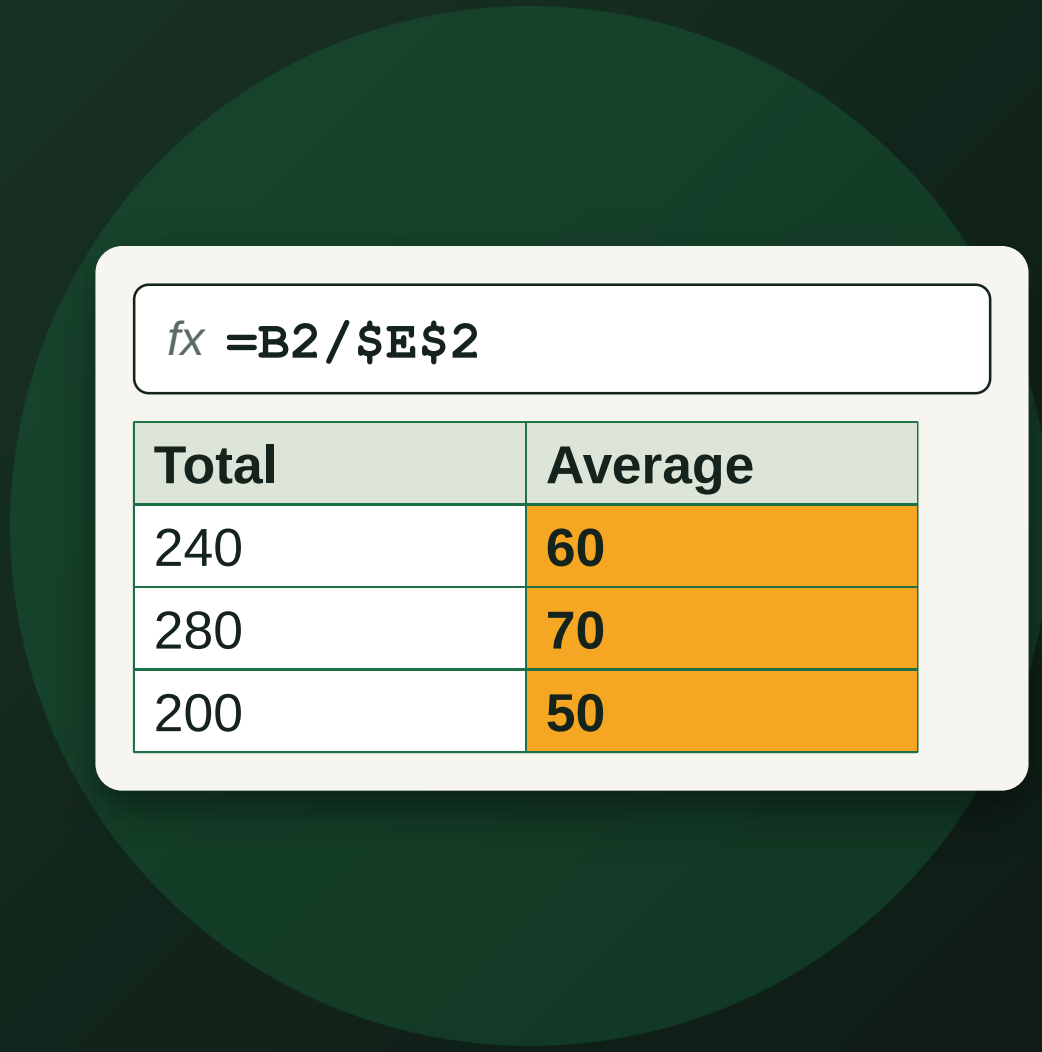
UNIT 10

Relative and Absolute References

`B2` `$E$2`

When we copy a formula, what should change and what should stay?



Slides by Sir M Khan Jamali

THEORY · THE KEY IDEA

What should move? What should stay?

C2 *fx* =B2/E2

	A	B	C	D	E
1	**Student**	**Total**	**Average**		**Subjects**
2	Ali	240	60		4
3	Sara	280			
4	Omar	200			

B2 · TOTAL · SHOULD MOVE

B2 → B3 → B4

Each student has their own total.

E2 · SUBJECTS · SHOULD STAY

E2 → E2 → E2

Everyone shares the same 4.

A **cell reference** is the name of a cell, like B2. Copying **moves** every reference, unless a **$** locks it.

EXAMPLE 1 OF 3

Relative references move

	A	B	C	D	E
1	**Student**	**Total**	**Average**		**Subjects**
2	Ali	240	60		4
3	Sara	280	#DIV/0!		
4	Omar	200	#DIV/0!		

COPY DOWN: BOTH MOVE

C2 `=B2/E2` → `60`

C3 `=B3/E3` → `#DIV/0!`

C4 `=B4/E4` → `#DIV/0!`

No $ signs, so **B** and **E** both move down. **E3** is empty, so Excel divides by zero.

EXAMPLE 2 OF 3

Absolute references stay

	A	B	C	D	E
1	**Student**	**Total**	**Average**		**Subjects**
2	Ali	240	60		4
3	Sara	280	70		
4	Omar	200	50		

COPY DOWN: $E$2 STAYS

C2 `=B2/$E$2` → 60

C3 `=B3/$E$2` → 70

C4 `=B4/$E$2` → 50

$E$2 locks column E and row 2, so every row divides by the same 4. Press **F4** to add the $ signs.

EXAMPLE 3 OF 3

Mixed references lock one part

D4 *fx* `=$A4*D$1`

	A	B	C	D	E
1		1	2	3	4
2	1	1	2	3	4
3	2	2	4	6	8
4	3	3	6	9	12
5	4	4	8	12	16

`$A4` Look **left**: the column is always A. The row follows the formula.

`D$1` Look **up**: the row is always 1. The column follows the formula.

Type **=$A2*B$1** in B2, then copy across and down. A is on the left, 1 is on top.

RECAP · THE PATTERN

Reading the dollar signs

BLUE = MOVES PURPLE = STAYS

B2	RELATIVE	Column and row both move.
$B$2	ABSOLUTE	Column and row both stay.
$B2	MIXED · COLUMN	Column stays, row moves.
B$2	MIXED · ROW	Row stays, column moves.

Key idea: relative moves, absolute stays. Press **F4** to cycle B2 → $B$2 → B$2 → $B2.

TROUBLESHOOTING

Why did my formula go wrong?

1 #DIV/0! or wrong answers below

No $ sign, so the shared cell moved down.

GOT `=B3/E3`

NEED `=B3/$E$2`

2 Every row gives the same answer

Too many $ signs: even B2 got locked.

GOT `=$B$2/$E$2`

NEED `=B3/$E$2`

3 #DIV/0! again after copying down

Wrong part locked: $E2 locks only the column.

GOT `=B3/$E3`

NEED `=B3/$E$2`

Tip: click the formula and press **F2**. Excel outlines the cells it looks at.

PRACTICE 1 OF 3 · ABSOLUTE REFERENCE

Exercise 1: bonus marks

	A	B	C	D	E	F
1	**Name**	**Marks**	**New Marks**		**Bonus**	**5**
2	Sara	70				
3	Ali	64				
4	Omar	81				
5	Hina	58				
6	Zoya	75				

Fill in: `New Marks: add` `Locked: F1`

Start with Sara: her marks are in B2 and the bonus is in F1.

Exit ticket: change F1 to 10. Do all the New Marks change?

PRACTICE 2 OF 3 · ABSOLUTE REFERENCE AND SUM

Exercise 2: class trip

	A	B	C	D	E	F
1	**Class**	**Students**	**Cost**		**Cost each**	**100**
2	6A	**28**				
3	6B	30				
4	7A	25				
5	7B	32				
6	8A	27				
7	**Total**					

Fill in: `Cost: multiply` `Locked: F1` `Total: SUM`

Start with 6A: its students are in B2 and the cost each is in F1.

Exit ticket: the cost each becomes 120. What is the new total cost?

PRACTICE 3 OF 3 · MIXED CHALLENGE

Exercise 3: pay table

	A	B	C	D	E
1	**Hours**	**10**	**15**	**20**	**Total pay**
2	2				
3	4				
4	6				
5	8				
6	**Total**				

Fill in: `Grid: multiply` `Total pay: SUM` `Total: SUM`

Start with B2: the hours are in column A and the rates are in row 1.

Exit ticket: which part of your formula is locked, and why?